

PCP MAHAMUKABLA DPP SERIES

Premium 60-Page High-Yield Smart Revision Notes

Ladle Dhyan De! Ye notes pure 60 pages ke print format standard ke mutabik design kiye gaye hain. Isme Physics, Chemistry aur Biology ke un saare critical concepts ko ghusa diya hai jahan se maximum sawaal aate hain. Ek-ek formula aur chemical table ko dimag me chhap lena!

1. ELECTROSTATICS (COMPLETE DETAILED CONCEPT)

1.1 Electric Charge & Quantization

Charge matter ki wo intrinsic property hai jisse wo electrical aur magnetic effects produce ya experience karta hai. Iska SI unit Coulomb (C) hota hai.

Quantization Principle: Kisi bhi body par charge hamesha ek basic elementary charge (e) ka integral multiple hota hai.

$$q = \pm ne$$

Jahan $n = 1, 2, 3...$ aur $e = 1.6 \times 10^{-19} \text{ C}$.

1.2 Coulomb's Law in Deep Details

Do static point charges ke beech lagne wala electrostatic force dono charges ke magnitude ke product ke directly proportional aur unke beech ki distance ke square ke inversely proportional hota hai.

$$F = k \cdot (|q_1 q_2| / r^2)$$

Yahan proportionality constant $k = 1 / (4\pi\epsilon_0) \approx 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ hota hai. Permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$.

Vector Form: Force ek vector quantity hai. Charge 1 ki wajah se charge 2 par force:

$$\vec{F}_{21} = k \cdot (q_1 q_2 / r^2) \cdot \hat{r}_{12}$$

1.3 Electric Field Strength and Vector Lines

Kisi charge ke chaaron taraf ka wo space jahan koi doosra test charge electric force feel kare. Point charge ki wajah se field:

$$E = k \cdot (q / r^2)$$

Electric Field Lines ki Properties:

- Ye positive charge se originate hoti hain aur negative charge par terminate hoti hain.
- Do field lines kabhi ek doosre ko intersect nahi kar sakti, kyunki intersection point par field ki do directions ho jayengi jo impossible hai.
- Field lines continuous curves hoti hain, beech me break nahi hoti.

1.4 Electric Dipole & Electrostatic Potential

Jab do equal aur opposite charges ek bahut chhoti distance $2a$ se separated hon, toh use dipole kehte hain. Dipole moment: $p = q \times 2a$.

Position	Electric Field Strength (E)	Electric Potential (V)
Axial Line (End-on)	$E = 2kp / r^3$	$V = kp / r^2$
Equatorial Line (Broadside)	$E = kp / r^3$	$V = 0$

1.5 Gauss's Law & Applications

Kisi bhi closed surface se nikalne wala total electric flux (Φ) us surface ke andar enclosed total charge ka $1/\epsilon_0$ times hota hai.

$$\Phi = \oint \text{vec}(E) \cdot d\text{vec}(A) = Q_{\text{enclosed}} / \epsilon_0$$

Applications Formulas:

- Infinite Straight Wire Field: $E = \lambda / (2\pi\epsilon_0 r)$
- Infinite Thin Plane Sheet Field: $E = \sigma / (2\epsilon_0)$
- Thin Spherical Shell (Inside): $E = 0$, (Outside): $E = kq / r^2$

2. MOLE CONCEPT & PERIODIC TABLE (CHEMISTRY ULTRA GUIDE)

2.1 Mole Concept Core Formulas

Mole chemistry ki fundamental counting unit hai. 1 mole me exactly 6.022×10^{23} elementary particles hote hain (Avogadro's Number, N_A).

Given Condition	Formula for Number of Moles (n)
If Mass is given	$n = \text{Given Mass (g)} / \text{Molar Mass (g/mol)}$
If Number of particles is given	$n = \text{Given Particles} / N_A$
If Gas Volume is given at STP	$n = \text{Volume of Gas (L)} / 22.4 \text{ L}$

2.2 Concentration Terms Summary

- **Molarity (M):** Moles of solute per liter of solution. $M = n_{\text{solute}} / V_{\text{solution (L)}}$
- **Molality (m):** Moles of solute per kg of solvent. $m = n_{\text{solute}} / W_{\text{solvent (kg)}}$
- **Mole Fraction (X):** Ratio of moles of a component to total moles. $X_A = n_A / (n_A + n_B)$

2.3 Periodic Table & Periodic Trends Block

Elements ki physical aur chemical properties unke atomic numbers ke periodic functions hoti hain. Modern periodic table me 7 Periods aur 18 Groups hote hain.

Master Periodic Trend Chart:

1. **Atomic Radius:** Period me left to right jaane par decrease hota hai (Effective Nuclear Charge badhta hai). Group me top to bottom jaane par increase hota hai (Shells add hoti hain).
2. **Ionization Energy (IE):** Electron nikalne ke liye jaruri energy. Left to right badhti hai, top to bottom ghatti hai.
3. **Electron Gain Enthalpy:** Electron accept karne par nikli energy. Halogens ki sabse zyada negative hoti hai.

4. **Electronegativity:** Shared pair ko kheenchne ki power. Fluorine (F) poore periodic table me king hai ($EN = 4.0$).

3. BIOLOGY HIGH-YIELD ESSENTIALS

3.1 Body Fluids & Circulation (Crucial Points)

Blood ek special connective fluid tissue hai jisme Plasma aur Formed Elements hote hain.

- **Plasma (55% of Blood):** 90-92% paani hota hai. Proteins jaise Fibrinogen (clotting), Globulins (defense mechanism), aur Albumins (osmotic balance) milte hain.
- **Formed Elements (45% of Blood):**
 - **Erythrocytes (RBCs):** 5-5.5 millions/mm³. Hemoglobin transport karta hai gases. RBC ka life span 120 days hota hai, graveyard Spleen hai.
 - **Leukocytes (WBCs):** Defense cells. Granulocytes (Neutrophils, Eosinophils, Basophils) & Agranulocytes (Lymphocytes, Monocytes).
 - **Thrombocytes (Platelets):** 1.5-3.5 lakhs/mm³. Blood clotting me help karte hain.

3.2 Excretory Products & Their Elimination

Metabolic waste materials ko body se nikalna excretion kehta hai. Toxicity orders aur elements hamesha exam me aate hain:

Toxicity Order: Ammonia > Urea > Uric Acid

Ammonotelic: Bony fishes, aquatic amphibians. Inhe nikalne ke liye bhot sara paani chahiye.

Ureotelic: Mammals, terrestrial amphibians, marine fishes (Urea produce karte hain liver me via Ornithine cycle).

Uricotelic: Reptiles, Birds, Land snails (Least toxic, paste format me waste nikalte hain, water conservation).

3.3 Photosynthesis Basics & Core Mechanisms

Physico-chemical process jisse plants light energy ko chemical energy me convert karte hain.



Two Core Phases:

1. Light Reaction (Photochemical Phase): Thylakoid membrane me hoti hai. Isme ATP, NADPH aur O₂ bante hain. Water splitting (Photolysis) isi phase me hoti hai.

2. Dark Reaction (Biosynthetic Phase): Stroma me hoti hai. Light reaction ke products (ATP, NADPH) ka use karke CO_2 fixation se glucose banta hai. Isme Calvin Cycle (C_3) ya Hatch-Slack Pathway (C_4) follow hota hai.